

Introduction to mechanistic ecological niche models

Day 1: Introduction

Urtzi Enriquez-Urzelai¹

¹Institute of Vertebrate Biology,
Czech Academy of Sciences

 [GitHub](#)  [webpage](#)

2026-06-08



Organized by:



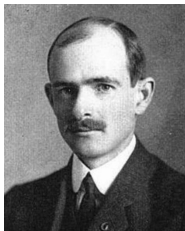
mechanistic ecological niche models

mechanistic ecological niche models

What is the ecological niche?

Grinnell (1917)

- niche as climatic and habitat requirements
- emphasis on abiotic conditions



Elton (1927)

- niche as the role of a species
- emphasis on biotic interactions

These various circumstances, which emphasize dependence upon cover, and adaptation in physical structure and temperament thereto, go to demonstrate the nature of the ultimate associational niche occupied by the California Thrasher. This is one of the minor niches which with their occupants all together make up the chaparral association. It is, of course, axiomatic that no two species regularly established in a single fauna have precisely the same niche relationships.

What is the ecological niche?

Grinnell (1917)

- niche as climatic and habitat requirements
- emphasis on abiotic conditions



Elton (1927)

- niche as the role of a species
- emphasis on biotic interactions

Hutchinson (1957)

- niche as an **n-dimensional hypervolume**
- defined in environmental space
- conditions allowing population persistence

"The niche is defined as an n-dimensional hypervolume"
— **Hutchinson (1957)**

BAM diagram

Geographical space

G

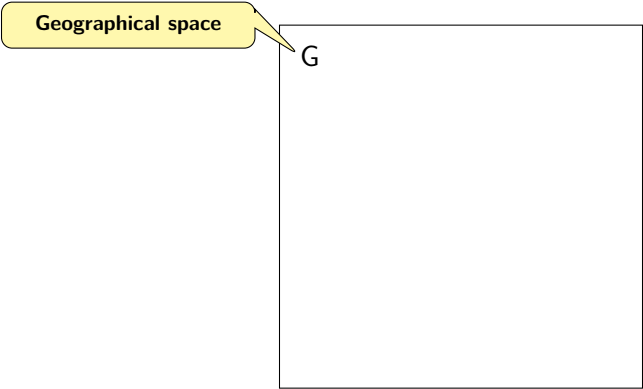


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

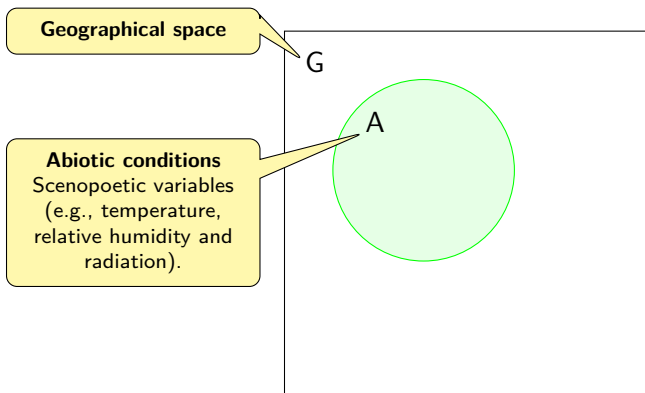


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

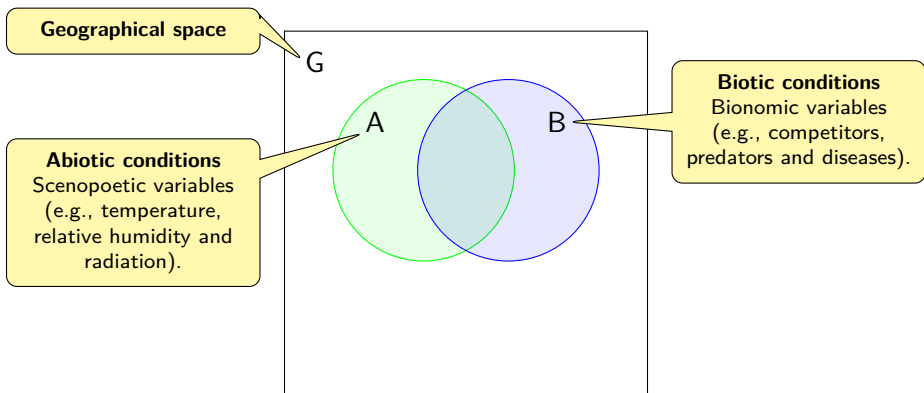


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

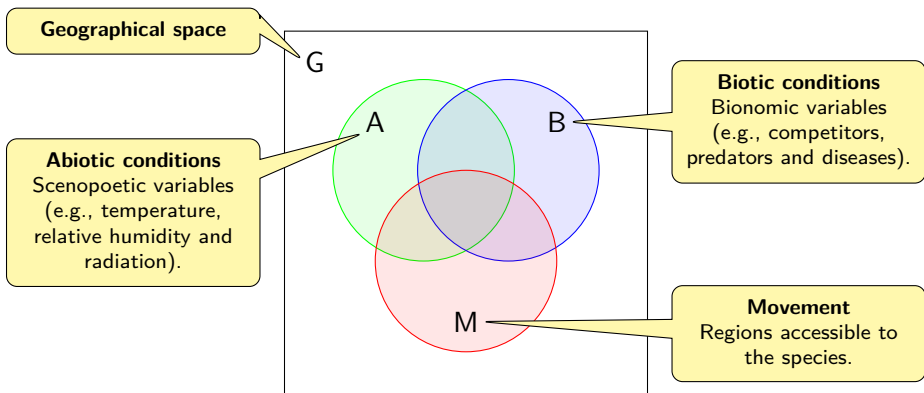


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

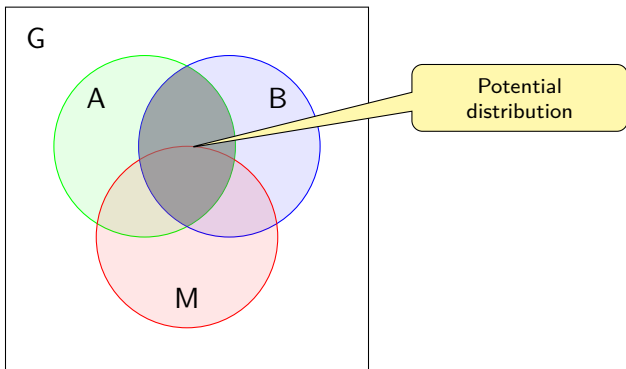


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

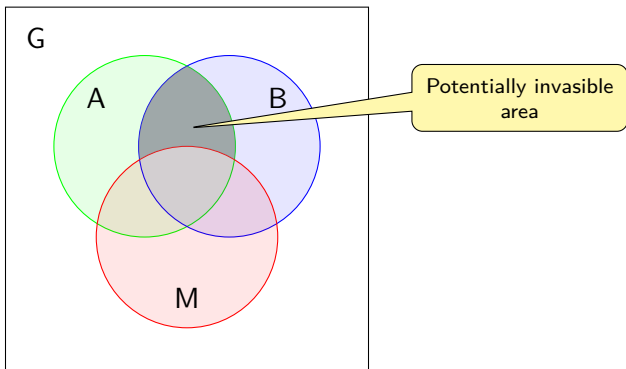


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

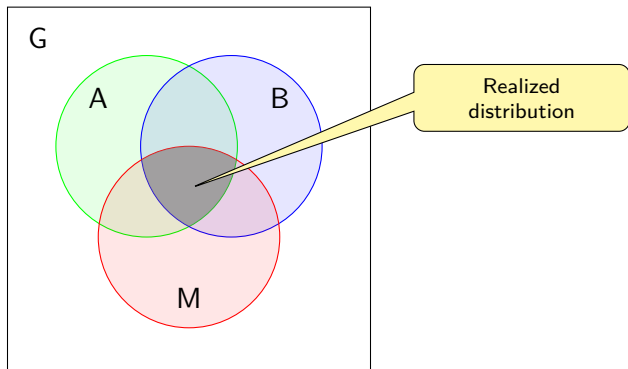


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

BAM diagram

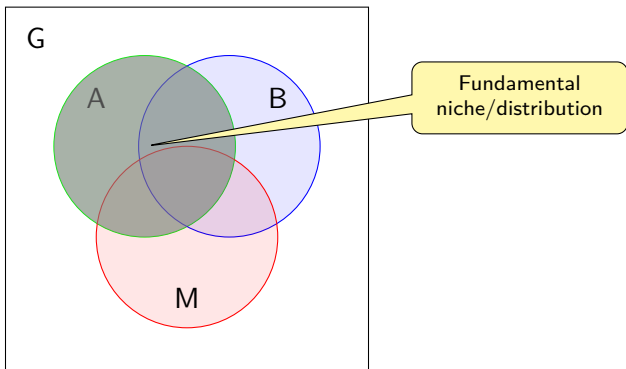
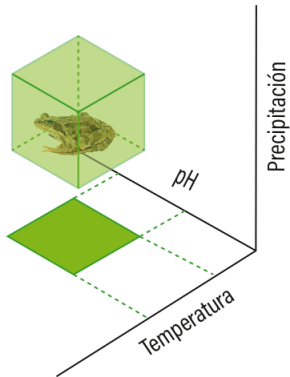
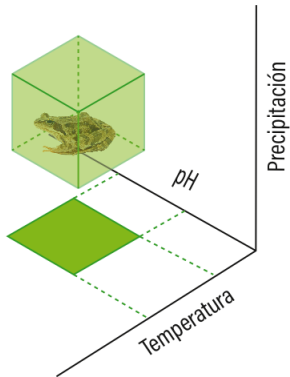


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

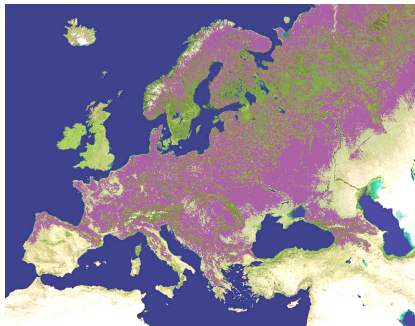
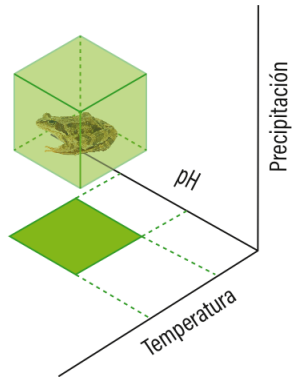
From niches to distributions



From niches to distributions



From niches to distributions



mechanistic ecological niche models

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

Dichotomies	
general	specific

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

Dichotomies	
general	specific
predictive	descriptive

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

Dichotomies	
general	specific
predictive	descriptive
process	pattern

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

Dichotomies	
general	specific
predictive	descriptive
process	pattern
mathematical	statistical

What are models

A model is a simplified representation of a real system, process, or phenomenon used to understand, explain, or predict its behavior.

Dichotomies	
general	specific
predictive	descriptive
process	pattern
mathematical	statistical
mechanistic	phenomenological

Modified from Bolker, [2008](#)

mechanistic ecological niche models

Mechanistic models are mathematical or computational representations that describe the underlying mechanisms of a system and its behavior by capturing the interactions and relationships between its components.

The Lotka-Volterra Model

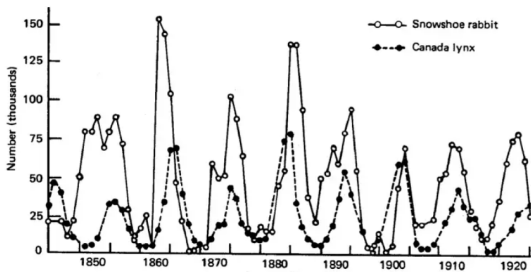


Figure 2: The classic population dynamics of hare and lynx

The Lotka-Volterra Model

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = \delta xy - \gamma y \end{cases}$$

Parameters:

x : density of prey (e.g., hare)

y : density of predators (e.g., lynx)

α : intrinsic growth rate of prey

β : predation rate

δ : predator reproduction rate

γ : predator mortality rate

The Lotka-Volterra Model

```
lv_model <- function(t, state, parameters) {  
  with(as.list(c(state, parameters)), {  
    dx <- alpha * x - beta * x * y  
    dy <- delta * x * y - gamma * y  
    list(c(dx, dy))  
  })  
}  
  
parameters <- c(  
  alpha = 0.6,  
  beta = 0.07,  
  delta = 0.04,  
  gamma = 0.9  
)  
  
state <- c(  
  x = lynxhare_df$Hare[1],  
  y = lynxhare_df$Lynx[1]  
)  
  
times <- seq(1, nrow(lynxhare_df), by = 1)  
  
out <- ode(  
  y = state,  
  times = times,  
  func = lv_model,  
  parms = parameters  
)
```

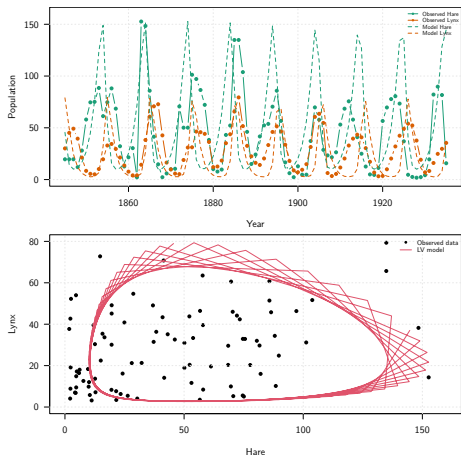


Figure 3: Observed hare-lynx dynamics and Lotka-Volterra simulation.

Models

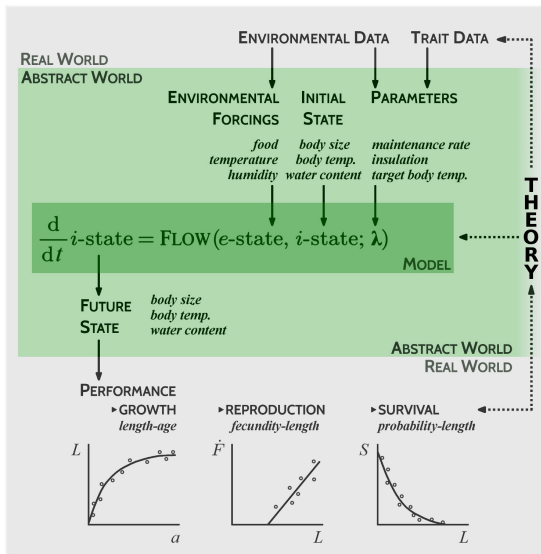


Figure 4: The links between the “real” and the “abstract” world, and how theory is used to connect them. Taken from Kearney et al. (2021)

mechanistic ecological niche models

Species Distribution Models (SDMs)



Environmental
layers

Figure 5: The two different philosophies to model species distributions. Correlative: where processes (or mechanisms) are implicit, and Mechanistic: where processes (or mechanisms) are explicitly included in the models.

Species Distribution Models (SDMs)

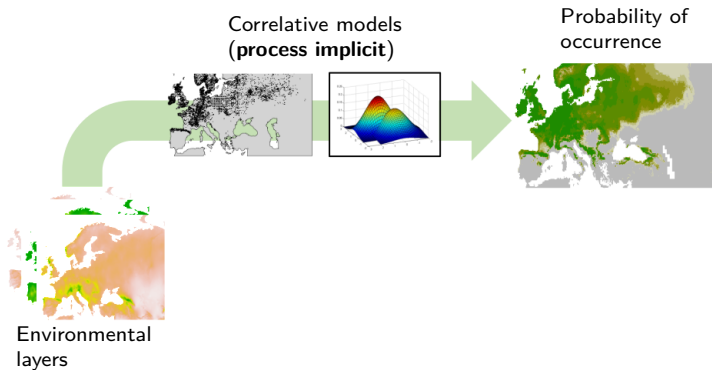


Figure 5: The two different philosophies to model species distributions. Correlative: where processes (or mechanisms) are implicit, and Mechanistic: where processes (or mechanisms) are explicitly included in the models.

Species Distribution Models (SDMs)

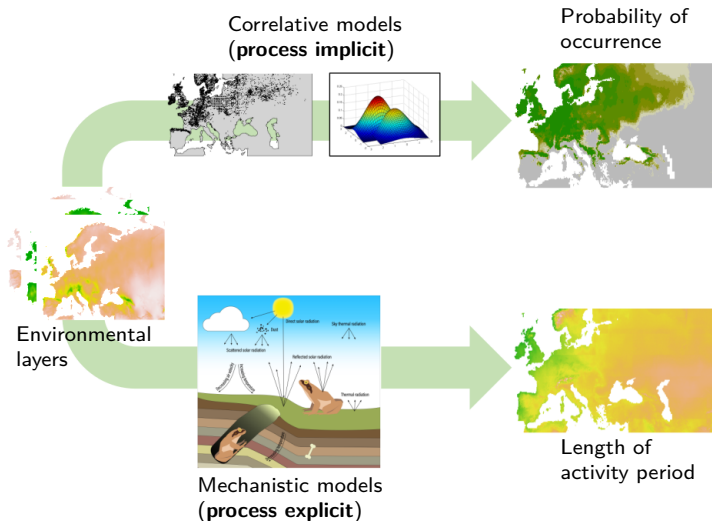


Figure 5: The two different philosophies to model species distributions. Correlative: where processes (or mechanisms) are implicit, and Mechanistic: where processes (or mechanisms) are explicitly included in the models.

Why mechanistic models?

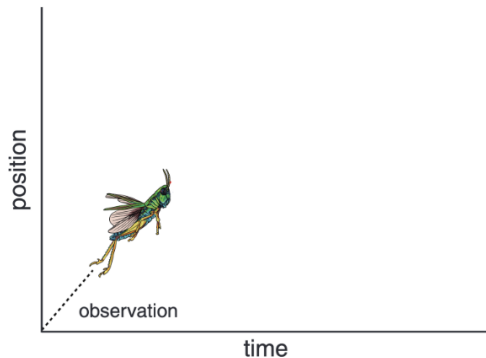


Figure 6: The problem (and solution) of extrapolation. Taken from Maino et al. (2016)

Why mechanistic models?

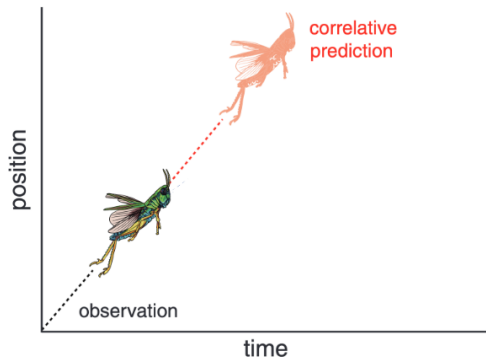


Figure 6: Taken from Maino et al. (2016)

Why mechanistic models?

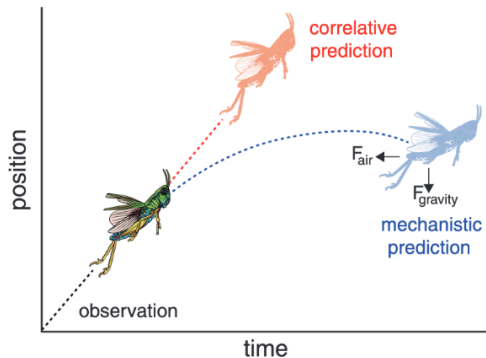


Figure 6: Taken from Maino et al. (2016)

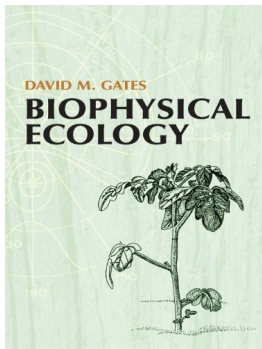
Standing on the shoulder of giants



David M. Gates



Warren P. Porter



Standing on the shoulder of giants



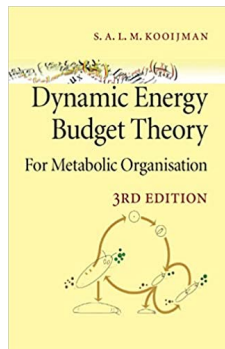
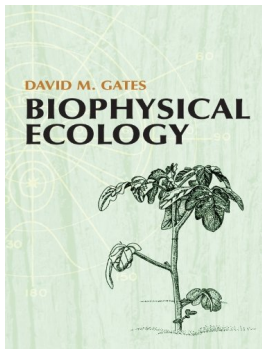
David M. Gates



Warren P. Porter



Bas Kooijman



References I

- Bolker, B. M. (2008). *Ecological models and data in R*. Princeton university press.
- Elton, C. S. (1927). *Animal ecology*. New York, Macmillan Co.
- Gates, D. M. (1980). *Biophysical ecology* (D. E. Reichle, Red.). Springer New York.
<https://doi.org/10.1007/978-1-4612-6024-0>
- Grinnell, J. (1917). The Niche-Relationships of the California Thrasher. *The Auk*, 34(4), 427–433. <https://doi.org/10.2307/4072271>
- Hutchinson, G. E. (1957). Concluding remarks. *Cold Spring Harbor Symposia on Quantitative Biology*, 22(0), 415–427. <https://doi.org/10.1101/SQB.1957.022.01.039>
- Kearney, M. R., Jusup, M., McGeoch, M. A., Kooijman, S. A. L. M., & Chown, S. L. (2021). Where do functional traits come from? The role of theory and models. *Functional Ecology*, 35(7), 1385–1396. <https://doi.org/10.1111/1365-2435.13829>
- Kooijman, S. A. L. M. (2010). *Dynamic energy budget theory for metabolic organisation* (3rd ed.). Cambridge university press.
- Maino, J. L., Kong, J. D., Hoffmann, A. A., Barton, M. G., & Kearney, M. R. (2016). Mechanistic models for predicting insect responses to climate change. *Current Opinion in Insect Science*, 17, 81–86. <https://doi.org/10.1016/j.cois.2016.07.006>

- Sillero, N. (2011). What does ecological modelling model? A proposed classification of ecological niche models based on their underlying methods. *Ecological Modelling*, 222(8), 1343–1346. <https://doi.org/10.1016/j.ecolmodel.2011.01.018>
- Soberón, J., & Nakamura, M. (2009). Niches and distributional areas: Concepts, methods, and assumptions. *Proceedings of the National Academy of Sciences*, 106, 19644–19650. <https://doi.org/10.1073/pnas.0901637106>